THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 1 Solutions

1. When $a = b = 1$, we have $\dfrac{2ab}{a^2+b^2} = 1$, so $1 \in S$. Thus, no upper bound for S can be less than 1. Also, we have
$$1 - \frac{2ab}{a^2+b^2} = \frac{a^2+b^2-2ab}{a^2+b^2} = \frac{(a-b)^2}{a^2+b^2} \geqslant 0$$
for any $a, b \in \mathbb{N}$. So 1 is an upper bound for S. These show $\sup S = 1$.

   Since $a, b > 0$, it is obvious that $\dfrac{2ab}{a^2+b^2} > 0$. So 0 is a lower bound for S. For any $\varepsilon > 0$, we choose $a \in \mathbb{N}$ such that $a > \dfrac{2}{\varepsilon}$ using the Archimedean property. For this a, we have
$$\frac{2a(1)}{a^2+1^2} = \frac{2a}{a^2+1} < \frac{2a}{a^2} = \frac{2}{a} < \varepsilon.$$
   This shows $\inf S = 0$ by Proposition 1.8.

2. Since $\{a_n\}$ converges, it is a Cauchy sequence by the Cauchy criterion. For any $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that $|a_m - a_n| < \dfrac{\varepsilon}{3}$ for any $m, n \geqslant N_1$. Define $N_2 = \max\left\{N_1, \left\lceil \dfrac{3}{\varepsilon} \right\rceil\right\}$. Then for any $m, n \geqslant N_2$, we have
$$\begin{aligned} |b_m - b_n| &= |(b_m - a_m) + (a_m - a_n) + (a_n - b_n)| \\ &\leqslant |b_m - a_m| + |a_m - a_n| + |a_n - b_n| \\ &< \frac{1}{m} + \frac{\varepsilon}{3} + \frac{1}{n} \\ &\leqslant \frac{1}{N_2} + \frac{\varepsilon}{3} + \frac{1}{N_2} \\ &\leqslant \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$
   This shows $\{b_n\}$ is a Cauchy sequence, and hence $\{b_n\}$ converges by the Cauchy criterion.

3. Note that
$$a_n - a_{n+1} = a_n - \frac{3a_n^2}{a_n^2+2} = \frac{a_n(a_n^2+2) - 3a_n^2}{a_n^2+2} = \frac{a_n(a_n-1)(a_n-2)}{a_n^2+2}.$$
   This implies $a_n = a_{n+1}$ if $a_n = 0, 1, 2$, $a_n > a_{n+1}$ if $a_n > 2$ or $0 < a_n < 1$, and $a_n < a_{n+1}$ if $1 < a_n < 2$. If $\{a_n\}$ converges to L, then $L - L = \dfrac{L(L-1)(L-2)}{L^2+2}$, and hence $L = 0, 1, 2$.

   - Case 1: $c = 0, 1, 2$
     From above, $\{a_n\}$ is a constant sequence, and the limit of $\{a_n\}$ is c.

- Case 2: $c > 2$

  Note that $a_2 < a_1$ and $a_2 = \dfrac{3a_1^2}{a_1^2+2} > 2$ (since this is equivalent to $a_1^2 > 4$). Inductively, we find that $a_{n+1} < a_n$ and $a_{n+1} > 2$ for any $n \in \mathbb{N}$. This shows $\{a_n\}$ is decreasing and bounded below by 2. Hence it is convergent by the monotone convergence theorem, and the limit L must be 2 since $L \geqslant 2$.

- Case 3: $0 < c < 1$

  Note that $a_2 < a_1$, and hence $0 < a_2 < 1$. Inductively, we find that $a_{n+1} < a_n$ and $0 < a_{n+1} < 1$ for any $n \in \mathbb{N}$. This shows $\{a_n\}$ is decreasing and bounded below by 0. Hence it is convergent by the monotone convergence theorem, and the limit L must be 0 since we must have $0 \leqslant L \leqslant c < 1$.

- Case 4: $1 < c < 2$

  Note that $a_2 > a_1$ and $a_2 = \dfrac{3a_1^2}{a_1^2+2} < 2$ (since this is equivalent to $a_1^2 < 4$). Inductively, we find that $a_{n+1} > a_n$ and $1 < a_{n+1} < 2$ for any $n \in \mathbb{N}$. This shows $\{a_n\}$ is increasing and bounded above by 2. Hence it is convergent by the monotone convergence theorem, and the limit L must be 2 since we must have $1 < c \leqslant L \leqslant 2$.

4. It is the same as proving $\{b_n\}$ is a bounded sequence, since the limit superior and the limit inferior of a bounded sequence must exist as real numbers. Note that $b_n > 0$ for any $n \in \mathbb{N}$.

   If $b_n \geqslant 1$, then $b_{n+1} > \dfrac{b_n}{2} \geqslant \dfrac{1}{2}$. If $b_n < 1$, then $b_{n+1} > \dfrac{1}{b_n^7} > 1$. Therefore, we must have $b_{n+1} > \dfrac{1}{2}$ in any case. So $\{b_n\}$ is bounded below by $\dfrac{1}{2}$.

   We claim that 256 is an upper bound for $\{b_n\}$. The base case $b_1 = 1 \leqslant 256$ clearly holds. Assuming $b_k \leqslant 256$ for some $k \in \mathbb{N}$, then

   $$b_{k+1} = \frac{b_k}{2} + \frac{1}{b_k^7} \leqslant \frac{256}{2} + \left(\frac{1}{2}\right)^{-7} = 256$$

   since $b_k \geqslant \dfrac{1}{2}$. This proves the claim by induction. So $\{b_n\}$ is bounded as desired.

5. (a) False. Define $\{a_n\}$ by $a_n = 1$ if n is a multiple of 3 and $a_n = -1$ otherwise. Then $\{a_{3n}\}$ is the constant 1 sequence, which converges to 1, while $\{a_{3n-1}\}$ and $\{a_{3n-2}\}$ are the constant -1 sequences, which converge to -1. But $\{a_n\}$ diverges.

   (b) True. Suppose $\{b_n\}$ converges to L. Since $\{b_{n^2}\}$ is a subsequence of $\{b_n\}$, it converges to L. So $\{c_n\}$ converges to $L - L = 0$.

   (c) False. Consider $X = \left\{-\dfrac{1}{n} : n \in \mathbb{N}\right\} \cup \{1\}$. Then $\sup X = 1$. However, $Y = \{-n : n \in \mathbb{N}\} \cup \{1\}$ has no infimum.